

Representations of $SL_2(\mathbb{F}_q)$

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- ① Basic Definitions
- ② Parabolic Induction
- ③ The Drinfeld Curve
- ④ Deligne-Lusztig Induction

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Definitions

Let $q > 3$ be an odd prime power.

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G has $q + 4$ conjugacy classes, so it has $q + 4$ irreducible representations.

Important Subgroups

- Define $B = \left\{ \begin{pmatrix} a & b \\ 0 & a^{-1} \end{pmatrix} \mid a \in \mathbb{F}_q^\times, b \in \mathbb{F}_q \right\} \subset G$.

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- Define $T = \left\{ \begin{pmatrix} a & 0 \\ 0 & a^{-1} \end{pmatrix} \mid a \in \mathbb{F}_q^\times \right\} \subset G$. Define $\mathbf{d}(a) = \begin{pmatrix} a & 0 \\ 0 & a^{-1} \end{pmatrix}$.

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- Define $U = \left\{ \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix} \mid x \in \mathbb{F}_q \right\} \subset G$. Let $\mathbf{u}(x) = \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix}$.

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Proposition

B can be written as the semidirect product $B = T \ltimes U$, where

$$\begin{pmatrix} a & b \\ 0 & a^{-1} \end{pmatrix} = \begin{pmatrix} a & 0 \\ 0 & a^{-1} \end{pmatrix} \begin{pmatrix} 1 & a^{-1}b \\ 0 & 1 \end{pmatrix}$$

Bruhat Decomposition

Define $s = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$, and observe $s^2 = -I$.

Proposition

G partitions into the double cosets of B as

$$G = B \sqcup BsB.$$

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Proof

For any $g = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in G \setminus B$, we have $c \neq 0$. Then

$$s^{-1}\mathbf{u}(-ac^{-1})g \in B.$$

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Induction

Since $T \cong \mathbb{F}_q^\times \cong \mu_{q-1}$, it has $q - 1$ linear characters.

Definition

Let α be a linear character of T . Define the character α_B on B as $\alpha_B(tu) = \alpha(t)$ for $t \in T, u \in U$. Let

$$R(\alpha) = \text{Ind}_B^G \alpha_B$$

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Each $R(\alpha)$ has degree $\deg R(\alpha) = [G : B] \cdot \deg \alpha_B = q + 1$.

Mackey Formula

Theorem

For any two linear characters α, β of T ,

$$\langle R(\alpha), R(\beta) \rangle_G = \langle \alpha, \beta \rangle_T + \langle \alpha, \beta^{-1} \rangle_T$$

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Proof Sketch

The Mackey formula on linear representation ρ of H is

$$\text{Res}_K^G \text{Ind}_H^G \rho \cong \bigoplus_{s \in K \backslash G/H} \text{Ind}_{K \cap {}^s H}^K \text{Res}_{K \cap {}^s H}^{{}^s H} {}^s \rho$$

Applied to G , this becomes $\text{Res}_B^G R(\beta) = \beta_B + \text{Ind}_T^B \text{Res}_T^B ({}^s \beta_B)$, and combining this with Frobenius reciprocity yields the desired equation, noting that $sts^{-1} = t^{-1}$ for all $t \in T$.

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- Let α_0 be the unique character of T of order 2. Then $\langle R(\alpha_0), R(\alpha_0) \rangle = 2$ so $R(\alpha_0) = R_+(\alpha_0) + R_-(\alpha_0)$. Both $R_+(\alpha_0)$ and $R_-(\alpha_0)$ have degree $(q+1)/2$.

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- If $\alpha^2 \neq 1$, then $\langle R(\alpha), R(\alpha) \rangle = \langle R(\alpha), R(\alpha^{-1}) \rangle = 1$ so $R(\alpha) = R(\alpha^{-1}) \in \text{Irr } G$.

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- If $\beta \notin \{\alpha, \alpha^{-1}\}$ then $\langle R(\alpha), R(\beta) \rangle = 0$ so all these characters are unique.

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Thus we have obtained $(q+5)/2$ irreducible characters of G .

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Let $\mathbb{F}$ be an algebraic closure of $\mathbb{F}_q$.

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This is an affine, smooth, irreducible curve.

Actions

Definition

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- The Frobenius endomorphism F sends $(x, y) \mapsto (x^q, y^q)$.

All three actions stabilize $\mathbf{Y}$ in $\mathbf{A}^2(\mathbb{F})$.

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- We have $F \circ \xi = \xi^{-1} \circ F$ for $\xi \in \mu_{q+1}$.

Thus we obtain an action of the monoid $G \times (\mu_{q+1} \rtimes \langle F \rangle)$ which stabilizes $\mathbf{Y}$.

Quotient

Definition

$$\mathbf{Y} = \{(x, y) \in \mathbf{A}^2(\mathbb{F}) \mid xy^q - yx^q = 1\}.$$

Proposition

The quotient $\mathbf{Y}/\mu_{q+1}$ is isomorphic to $\mathbf{P}^1(\mathbb{F}) \setminus \mathbf{P}^1(\mathbb{F}_q)$.

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Proof sketch

Consider the morphism $\pi : \mathbf{Y} \rightarrow \mathbf{P}^1(\mathbb{F}) \setminus \mathbf{P}^1(\mathbb{F}_q)$ given by $(x, y) \mapsto [x : y]$. This is a surjective map, and G acts on the right side also by matrix multiplication, which makes it a $G \times \langle F \rangle$ -equivariant morphism. We check that it is constant on μ_{q+1} -orbits.

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Properties:

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- Since $\dim \mathbf{Y} = 1$, these groups are nonzero only for $i = 1, 2$.

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Properties:

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- Since $\dim \mathbf{Y} = 1$, these groups are nonzero only for $i = 1, 2$.

Definition

Let θ be a character of μ_{q+1} and V_θ a $K\mu_{q+1}$ -module with character θ . There is a virtual character of G

$$R'(\theta) = \llbracket H_c^1(\mathbf{Y}) \otimes_{K\mu_{q+1}} V_\theta \rrbracket_G - \llbracket H_c^2(\mathbf{Y}) \otimes_{K\mu_{q+1}} V_\theta \rrbracket_G.$$

Introduction

- Since $\mathbf{Y}$ is irreducible, the top cohomology group $H_c^2(\mathbf{Y})$ has dimension 1 over K . Then the character $[[H_c^2(\mathbf{Y})]]_{G \times \mu_{q+1}}$ is the trivial character.

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- Since $\mathbf{Y}$ is irreducible, the top cohomology group $H_c^2(\mathbf{Y})$ has dimension 1 over K . Then the character $[[H_c^2(\mathbf{Y})]]_{G \times \mu_{q+1}}$ is the trivial character.
- Thus if θ is nontrivial, since the tensor product of a trivial and nontrivial representation is zero, the second term in $R'(\theta)$ disappears. Then

$$R'(\theta) = [[H_c^1(\mathbf{Y}) \otimes_{K\mu_{q+1}} V_\theta]]_G$$

and it is a character of G .

$R'(1)$

By definition,

$$\begin{aligned} R'(1) &= \llbracket H_c^1(\mathbf{Y}) \otimes_{K\mu_{q+1}} V_1 \rrbracket_G - \llbracket H_c^2(\mathbf{Y}) \otimes_{K\mu_{q+1}} V_1 \rrbracket_G \\ &= \llbracket H_c^1(\mathbf{Y})^{\mu_{q+1}} \rrbracket_G - \llbracket H_c^2(\mathbf{Y})^{\mu_{q+1}} \rrbracket_G. \end{aligned}$$

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We have $H_c^i(\mathbf{Y})^{\mu_{q+1}} \simeq H_c^i(\mathbf{Y}/\mu_{q+1})$, and by the previous section, $\mathbf{Y}/\mu_{q+1} \simeq \mathbf{P}^1(\mathbb{F}) \setminus \mathbf{P}^1(\mathbb{F}_q)$. Using some facts about the cohomology groups of the projective line, we obtain $R'(1) = \text{St}_G - 1_G$.

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- Furthermore, for all characters α of T , we have

$$\langle R(\alpha), R'(\theta) \rangle_G = 0.$$

Thus the irreducible characters obtained through Deligne-Lusztig induction are different from those obtained previously.

Mackey Formula

Theorem

Let θ, η be virtual characters of μ_{q+1} . Then

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- If $\theta^2 \neq 1$, the above yields $\langle R'(\theta), R'(\theta) \rangle = 1$. So $R'(\theta)$ is an irreducible character of G .

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- If η is not θ or $\theta^* = \theta^{-1}$, $\langle R'(\theta), R'(\eta) \rangle = 0$. Thus they are distinct.
- There is a unique character θ_0 of μ_{q+1} of order 2. $R'(\theta_0)$ is the sum of two irreducible characters, which we can calculate to both have degree $\frac{q-1}{2}$.

Conclusion

Through parabolic induction, we obtained $\frac{q+5}{2}$ irreducible characters of G .
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Through parabolic induction, we obtained $\frac{q+5}{2}$ irreducible characters of G . With Deligne-Lusztig induction, we obtained $\frac{q+3}{2}$ more irreducible characters:

- 2 from $R'(\theta_0)$
- 1 from each pair of θ, θ^{-1} where $\theta \neq \theta^{-1}$, which yields $\frac{q-1}{2}$.

Since G has $q + 4$ conjugacy classes, these are all the irreducible characters of G .

References

Cédric Bonnafé, *Representations of $\mathrm{SL}_2 \mathbb{F}_q$* , Algebra and Applications, Springer London, 2010

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